Milestone 2

Introduction:

In this project, the aim is to get the Spider Text-to-SQL Dataset ready for potential future use in machine learning and NLP projects. The Spider dataset contains several databases in the SQLite format, with different real-life scenarios and corresponding natural language questions as well as SQL queries.

However, before we apply any machine learning algorithm to the dataset, we need to comprehend its features and quality. Through Exploratory Data Analysis (EDA), we learn about database schemas, attribute distributions, data quality problems, etc. In this milestone, we focus on the collection of the dataset and its database profiling, and metadata generation.

Objective of Milestone 2:

The objectives of this milestone are to:

- Identify and acquire the required datasets
- Understand the database schemas and data characteristics
- Perform exploratory data analysis (EDA)
- Assess data quality and identify missing or inconsistent values
- Generate metadata and profiling reports
- Produce a training-ready dataset with documented preprocessing steps

Dataset Identification:

Spider: A Large-Scale Human-Labeled Dataset for Complex and Cross-Domain Semantic Parsing and Text-to-SQL.
Source:
Dataset Link: https://yale-lily.github.io/spider
GitHub Repository: https://github.com/taoyds/spider

License Status: The Spider dataset is publicly available for research purposes under its published licensing terms.

Purpose:

The dataset is designed for training and evaluating Text-to-SQL systems across multiple domains.

The main characteristics of the dataset include:

- Multiple real-world relational database domains
- Complex SQL query structures
- Cross-domain database schemas
- Human-generated natural language questions
- Support for evaluating generalization capabilities of Text-to-SQL models

Alternate Dataset:

Alternative datasets such as WikiSQL and CoSQL were considered.

Dataset Description:

The Spider dataset contains:

- Approximately 200 SQLite databases.
- More than 130 database domains.
- Thousands of tables and columns.
- Natural language questions paired with SQL queries.

Features

The dataset includes:

- Database tables
- Column names
- Data types
- Foreign key relationships
- Primary keys
- SQL queries
- Natural language questions

Target Variables

The target output is the SQL query corresponding to a natural language question.

Data Formats

- SQLite (.sqlite)
- JSON (.json)
- CSV (generated during preprocessing)

Sample Records:

Example database:

- academic.sqlite
- college_2.sqlite
- flight_2.sqlite

Dataset Schema

Each database consists of multiple relational tables linked through primary and foreign keys. Schema information was extracted automatically and exported into structured JSON and CSV files.

- JSON schema files
- Database summary CSV files
- Column profiling CSV files

The extracted metadata provides information about:

- Number of tables
- Number of columns
- Column data types
- Primary keys
- Foreign key relationships
- Attribute characteristics

Data Governance:

Data Source and Licensing

The dataset was obtained from the official Spider project repository. It is publicly available for academic and research use.

Privacy

The dataset does not contain personally identifiable information (PII). The databases are synthetic or anonymized and are intended for research purposes.

Data Quality

The dataset was validated by:

- Verifying SQLite database integrity.
- Profiling all tables and columns.
- Identifying missing values.
- Counting distinct values.
- Detecting unsupported or inconsistent encodings.

Ethics and Bias

Potential biases include:

- Uneven representation across application domains.
- SQL complexity variations.
- Limited coverage of certain industries.

The dataset is intended solely for research and educational purposes.

Reproducibility and Compliance

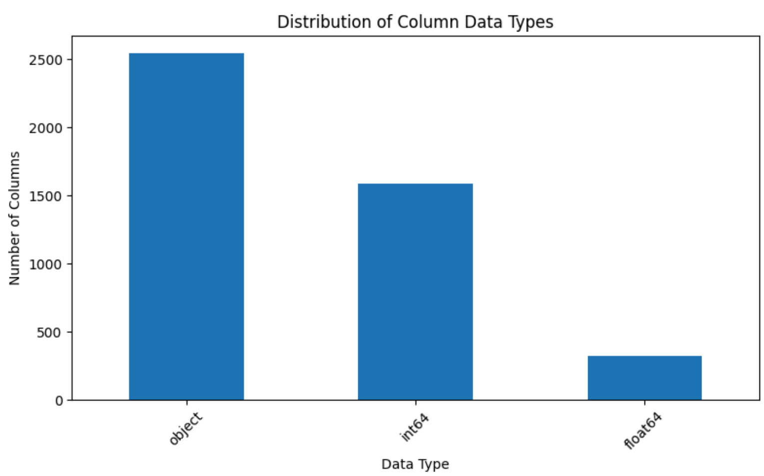
All preprocessing scripts were documented and version-controlled. The generated datasets can be reproduced by executing the provided Python scripts on the original Spider dataset.

Exploratory Data Analysis (EDA)

EDA was performed to understand the structural properties and quality of the Spider databases.

Data Type Distribution Analysis:

The distribution of attribute data types was analyzed to understand the composition of database columns.



Missing value analysis:

Missing value analysis was performed to identify incomplete attributes and potential data quality concerns.

The analysis provides:

- Columns containing missing values
- Number of missing entries
- Percentage of incomplete records

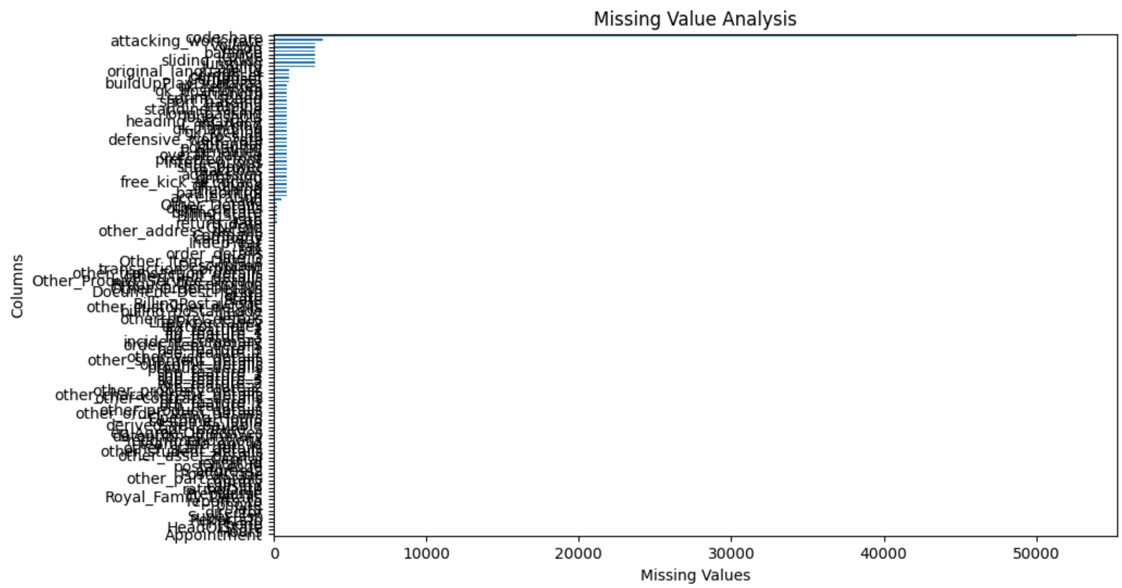
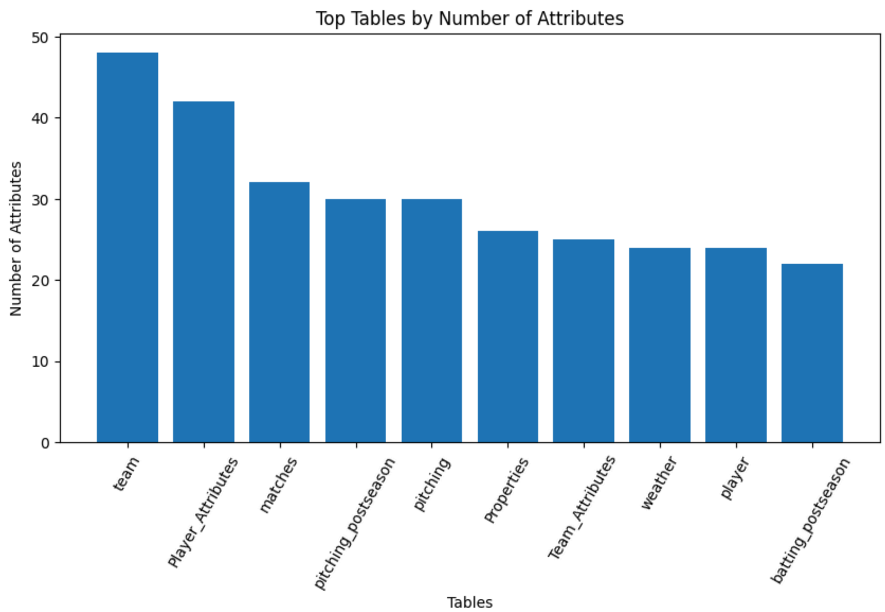


Table complexity analysis:

The number of attributes in each table was analyzed to identify complex database structures.

Tables with a larger number of columns indicate higher schema complexity.

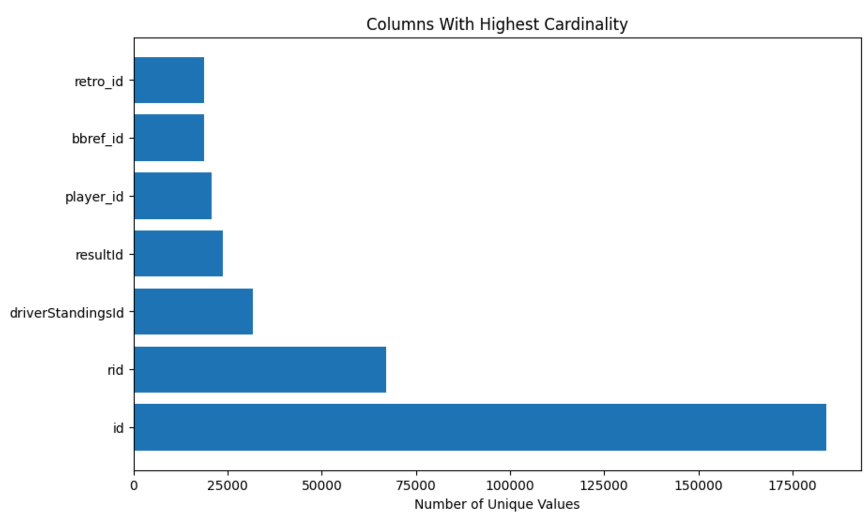


Attribute Cardinality Analysis:

Distinct value analysis was performed to identify columns with high uniqueness.

High-cardinality attributes often represent:

- Primary identifiers
- Transaction identifiers
- Unique entities



Data Preprocessing

The following preprocessing steps were completed:

- Automated database discovery.
- Schema extraction.
- Table profiling.
- Missing value detection.
- Data type identification.
- Removal of unreadable records caused by invalid text encodings (where applicable).
- Standardized export to CSV format.
- Metadata generation.

Since this milestone focuses on database profiling rather than predictive modeling, normalization, feature scaling, and categorical encoding were not required.

Dataset Integration

Multiple SQLite databases were processed independently.

No direct merging was performed because each database represents an independent application domain with its own schema.

Schema consistency was maintained by generating standardized metadata across all databases.

Dataset Splitting

The Spider dataset already provides predefined training and development splits.

The official train and development partitions were retained to ensure consistency with published benchmark evaluations.

Final Prepared Dataset

The final prepared dataset consists of:

- Original SQLite databases
- Extracted schema metadata
- Database summary reports
- Column profiling reports
- Cleaned metadata exports

The dataset is now organized and documented, enabling direct use in future machine learning experiments.

Challenges Encountered

The following challenges were encountered during preprocessing:

- Managing hundreds of SQLite databases
- Handling varying database schemas
- Processing inconsistent text encodings in certain databases
- Automating metadata extraction across diverse domains
- Generating consistent profiling outputs

Deliverables Produced

The following deliverables were generated:

- Database summary report (database_summary.csv)
- Column profiling report (column_summary.csv)
- Schema metadata (schema.json)
- Automated preprocessing scripts
- Documentation of preprocessing workflow
- Training-ready database collection

Summary

With this step, the Spider dataset was efficiently set up for any machine learning tasks in the future. Metadata collection and automated profiling have been completed for the SQLite databases, where summaries about the characteristics and data quality were created.

Key Observations

- The dataset is composed of several applications, each with its own schema.
- The sizes of tables and the number of columns differ greatly between databases.
- The majority of databases show excellent structural consistency, except for a few that have text encoding problems.

Planned Activities for Milestone 3

- Feature engineering for Text-to-SQL tasks.
- Training baseline Text-to-SQL models.
- Model evaluation using official Spider benchmarks.
- Error analysis and performance optimization.